

Yugandhar Narravula

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Education

Texas Tech University, Lubbock

Aug 2024 – May 2026

Master of Science in Computer Science

Relevant Coursework: Machine Learning, Neural Networks, Advanced Data Science, Analysis of Algorithms, Software Modeling Architecture

Work Experience

Texas Tech University — Research Assistant

Aug 2024 – May 2026

- Developed **AI/ML pipelines** for **data preprocessing**, **feature engineering**, and **model training** using **Python**, supporting research-driven **deep learning** and **NLP** experiments
- Implemented **LLM-based applied AI prototypes** for research workflows, including **prompt design**, **inference evaluation**, and **performance benchmarking** across multiple model architectures
- Automated **experiment tracking**, **model evaluation**, and **data processing** workflows to improve reproducibility, scalability, and research efficiency

Cognizant — Programmer Analyst Trainee

Jan 2024 – Jun 2024

- Developed **LLM-powered Generative AI** solutions using **prompt engineering** and enterprise data integration, improving response accuracy, relevance, and quality
- Built **RAG-based pipelines** by indexing and retrieving structured and unstructured enterprise data with **Azure Cognitive Search** for **context-grounded AI** applications
- Designed and maintained **REST APIs** and **backend services** to enable scalable integration of **GenAI** and **LLM inference** capabilities into enterprise applications

Projects

Enterprise AI Copilot for Document Intelligence - RAG, LLM, Re-Ranking — Live Demo

Tech Stack: Python, FastAPI, Streamlit, Qdrant, Sentence Transformers, LLaMA 3 (Groq)

- Designed a **3-tier RAG system** (**Streamlit Cloud** → **FastAPI on Render** → **Qdrant Cloud**) with **hybrid BM25 + vector retrieval** and **cross-encoder reranking** (ms-marco-MiniLM-L-6-v2) for precision document Q&A
- Integrated **Groq LLaMA 3.3-70B** with **NOT_FOUND grounding enforcement**, a keyword-based **query router** for web vs. RAG routing, and **three-level fallback logic** to eliminate unsupported answers
- Reduced redundant LLM inference with **Upstash Redis caching** (1-hour TTL) and implemented **heuristic confidence scoring**, deployed end-to-end across **Render**, **Streamlit Cloud**, and **Qdrant Cloud**

Enterprise Multi-Agent AI Copilot

Tech Stack: Python, LLMs, LangGraph, FastAPI, Streamlit, REST APIs, Multi-Agent Systems

- Architected a **LangGraph-based multi-agent system** with **11 specialized agents** and a shared **AtlasState TypedDict**, generating personalized 3–7 day travel itineraries in **under 10 seconds**
- Integrated **6 real-time APIs** (weather, geocoding, currency, attractions) with **async parallel execution**, reducing end-to-end latency by **61% (18s → 7s)**
- Deployed a **FastAPI** backend and **Streamlit** frontend with **conditional edge routing**, error recovery, and structured agent outputs across the full **LangGraph** pipeline

AI-Driven Portfolio Optimization & Time-Series Forecasting System — Live Demo

Tech Stack: Python, Prophet, SciPy, Supabase, Streamlit, CircleCI

- Built a fully automated daily **ML pipeline** using **Prophet** and **SciPy SLSQP (Modern Portfolio Theory)** to forecast stock prices and optimize a **12-ticker portfolio** via **GitHub Actions**, with results stored in **Supabase**
- Configured **Prophet** with **pandas-market-calendars** (12 US trading holidays) and **SciPy SLSQP** with $\lambda=5$ **risk-aversion** and **5–100% per-asset weight bounds** for constrained **mean-variance optimization**
- Implemented **CI/CD** with **CircleCI** parallel stages (lint, type check, pytest) and deployed a live **Streamlit** dashboard on **Streamlit Community Cloud** for real-time portfolio tracking

Technical Skills

Languages: Python, SQL, Java

AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, LLM Applications, Enterprise AI Agents, Secure AI Systems, Responsible AI, Model Training, Model Evaluation, Benchmarking, Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering, Multi-Agent Systems, Re-Ranking, Feature Engineering, Time-Series Forecasting

ML Frameworks / Tools: FastAPI, Streamlit, PyTorch, Scikit-learn, Sentence Transformers, LangChain, Scipy

LLM / GenAI Tools: LLaMA 3, Groq API, Embedding Models, Azure AI, Qdrant, LangGraph, Source Grounding, Enterprise Data Integration, OpenAI GPT-4o, GPT4All, Azure AI, vector Search, Cross-Encoder Reranking, BM25

Backend / Distributed Systems: REST APIs, Distributed Services, Microservices, Data Pipelines, PostgreSQL, Supabase, Asynchronous Processing, Production Monitoring, Zero Trust Principles, Compliance-Aware AI, Redis, Vector Databases

Deployment / MLOps: Docker, CI/CD, GitHub Actions, CircleCI, Cloud Deployment, Model Serving, Experiment Tracking

Developer Tools: Git, GitHub, Poetry, PyMuPDF, Yfinance